Haoyun HU

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EDUCATION

University of California, Berkeley – Haas School of Business | *Master of Financial Engineering* **Mar 2026 - Mar 2027**
The Chinese University of Hong Kong | *Bachelor's Degree in Financial Engineering* **Sept 2021 - May 2025**
Achievements: Tier One Scholarship for Academic Year 2022-2023, 2023-2024 (rank 1/176, **the highest scholarship**, **11,600 USD** per year)

AWARDS & SKILLS

Awards: Top 1%, Honor Roll of Distinction Certificate in 2018 American Mathematics Competition, Second prize in 2018 National High School Mathematics Competition
Mathematics: Calculus, Linear Algebra, Probability and Statistics, Stochastic Processes, Econometrics, Stochastic Calculus, Mathematical Analysis, Numerical Analysis, ODE, Discrete Mathematics
Programming: Python (Pandas, NumPy, Statsmodels, SciPy, PyTorch, skorch), C++, Data Structures, Machine Learning
Professional Texas Hold'em poker: Managed a book of over $200K for a family office, with cumulative profits $300K+ in less than 6 months

PROFESSIONAL EXPERIENCE

Ultramarin | *Quantitative Software Engineer Intern (Research-Focused) - Equity Selection Team* **Berlin, Germany |** *05/2026 – 08/2026*

- Ensemble Model Development: Designed an Equal Risk Contribution ensemble model for equity alpha forecasting from multiple base models using SLSQP solver (SciPy), introducing a Ledoit-Wolf shrinkage parameter to stabilize optimization and improve robustness against estimator redundancy and variance domination
- Linear-Guided Neural Network (LGNN) Implementation: Incorporated Fama–MacBeth Ridge regression as a frozen teacher to guide an MLP (torch) by distilling stable linear factor structure, to mitigate overfitting and enhance out-of-sample generalization
- LGNN Experimental Validation: Performed systematic hyperparameter tuning on controlled synthetic datasets under known target structures, analyzing learning behavior, training stability, and out-of-range extrapolation through partial dependency plots
- Alpha Research and Enhancement: Developed new dynamic metrics like signal decay and turnover, and improved the temporal stability of some trend and reversal factors by 20% using methods like EWMA smoothing and exclusion windows

Super Quantum Capital Management (AUM: 2B USD) | *Quantitative Research Intern* **Shenzhen, China |** *10/2025 – 03/2026*

- Asset Allocation: Constructed factor-mimicking portfolios using Python (Pandas, NumPy, Statsmodels) across 18 Chinese and US asset classes with 200+ macro factors
- Portfolio Optimization and Attribution Analysis: Developed a risk parity optimization framework with constrained SLSQP solver (SciPy) and decomposed portfolio risk exposure across macro factors, achieving a Sharpe ratio of 1.47 over the past 10 years
- Object-oriented Python Implementation: Encapsulated the asset allocation workflow into a reusable Python module, integrating 30+ core functionalities including data preprocessing, factor development, risk parity optimization, and portfolio evaluation

Ricequant | *Quantitative Research Intern* **Shenzhen, China |** *05/2024 – 08/2024*

- Forecast Model Construction: Aligned factor values with the latest financial reporting cycles using analyst forecasts and 10,000+ financial data points to identify seasonality and stability of quarterly earnings, improving ICIR by 0.08-0.1 for each factor
- Alternative Information Extraction: Extracted insights and optimized factors from alternative financial data including financial statement notes and analyst forecasts, constructing 30+ static and dynamic factors with ICIR > 0.4 (Pandas, NumPy, Statsmodels)
- Barra Model Development: Estimated covariance matrices for 500+ mainland equities with Barra China Equity Model factors, improving risk prediction accuracy by 15% through Newey-West adjustment, structural modeling, and Bayesian shrinkage

RESEARCH EXPERIENCE

Goldman Sachs | Industry Project, Predictive Model for the US Power Market **Berkeley, CA |** *10/2026 – 12/2026*

- Power Price Forecasting: Developing an hourly ERCOT electricity price forecasting framework with machine learning methods and forward-price calibration, using demand, wind and solar generation data to model net-load dynamics and intraday price patterns

NewMark Risk | Industry Project, *Structural Effects of Derivative Expirations* **Berkeley, CA |** *03/2026 – 05/2026*

- Equity Derivatives Microstructure Analysis: Statistically tested the expiration-related "pinning" effect and identified systematic price distortions and volatility shifts in HKEX, providing empirical insights into delta hedging behavior and market microstructure

The Chinese University of Hong Kong, Shenzhen **|** ***Research Assistant (Poker Strategy Exploration)*** **Shenzhen, China |** *07/2025 – 10/2025*

- Game-Theoretic Algorithm Development: Implemented a counterfactual regret minimization (CFR) method to obtain the Nash equilibrium solution for Leduc Hold'em poker game, improving the expected payoff of the game by 66%
- Poker Feature Engineering: Cooperated with the AI team of GTO Wizard to construct 20+ human-understandable poker features for Texas Hold'em poker, like blockers and range features, and explored their effects on per-hand strategies